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$$\begin{aligned}\int_1^{+\infty} \frac{\ln(x)}{x^2} dx &= - \int_1^{+\infty} \ln(x) d\left(\frac{1}{x}\right) = \lim_{a \rightarrow +\infty} - \int_1^a \ln(x) d\left(\frac{1}{x}\right) \\ &= \lim_{a \rightarrow +\infty} \left[\frac{\ln(x)}{x} \right]_1^a - \lim_{a \rightarrow +\infty} \left[\frac{1}{x} \right]_1^a = -0 + 0 - 0 + 1 = 1\end{aligned}$$

References